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The interplay of riverscape features and exotic introgression on the genetic structure of the Mexican golden trout (*Oncorhynchus chrysogaster*), a simulation approach

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Abstract

Aim: We explored the factors between riverscape features and exotic introgression of cultured rainbow trout (*Oncorhynchus mykiss*) that drive the current genetic structure in the endemic Mexican golden trout (*Oncorhynchus chrysogaster*) using a riverscape genetics simulator.

Location: Sierra Madre Occidental, Northwest Mexico, North America.

Methods: We created a system model for Mexican golden trout to test various scenarios for the effect of riverscape resistance and rainbow trout introgression on the native genetic structure of the Mexican golden trout. Individual genotypes were used to initialize the model and were based upon empirical allele frequencies from three genetically and spatially distinct groups in the study area. The geographical location of the simulated populations was defined using a species distribution model. We included hypotheses for movement of riverine distance and riverscape resistance. We evaluated model performance by conducting population and landscape genetics analyses on the simulation outputs.

Results: The most plausible scenario included riverscape resistance without introgression, which presented a spatial genetic structure matching the observed riverscape genetic structure and fitting expected general life-history patterns for high elevation trout streams. In addition, temperature, precipitation, hydrology and topography were the most important variables influencing species occurrence and dispersal.

Main conclusion: This work presents one of the first empirically derived simulation studies of introgression. It highlights the importance of considering riverscape resistance (i.e. hydroclimatic and topographical gradient effects) in models of spatial genetic structure and introgression. Moreover, our modelling suggests that what we characterize as riverscape resistance is acting as a boundary against exotic introgression. The results of this study could help to develop management strategies that preserve native gene pools.

KEYWORDS

CDMetaPOP, exotic trout, *Oncorhynchus chrysogaster*, rainbow trout, riverscape genetics, species distribution modelling

1 | INTRODUCTION

Genetic diversity is the result of various micro-evolutionary processes (mutation, gene flow, genetic drift and natural selection) that tend to produce local genetic differentiation among populations (King, Stansfield, & Mulligan, 2006). This genetic diversity allows species to adapt and persist in changing environments, and is considered an important element of biodiversity management (Allendorf, Hohenlohe, & Luikart, 2010).

Demographics have been correlated with variation in genetic diversity (King et al., 2006). For instance, if populations are established with a small number of individuals and receive little immigration (i.e. low gene flow) from closely related populations, changes in demographic processes (e.g. reproduction, fecundity, egg and larvae survival) resulting in an increase in mortality could cause a loss of genetic diversity (Banks et al., 2013). Furthermore, the loss of alleles could lead to a decrease in fitness (selection-driven processes), reducing skills to cope with changing environments (Kokko et al., 2017).

Micro-evolutionary processes can also be affected at different environmental scales. For example, climatic patterns can have an impact on population sizes and landscape structure could prevent gene flow (Manel & Holderegger, 2013). Therefore, evaluating scale-dependent environmental impacts on overall genetic diversity and structure is important for our understanding and promotion of conservation and the effective management of genetic diversity for endangered populations (Allendorf et al., 2010).

Aquatic organisms inhabiting riverine environments depend on hierarchical systems influenced by climate, topography and land use at multiple scales. These factors drive processes that divide environments into microhabitats (Allen & Starr, 1982; Manel, Dias, Buckton, & Ormerod, 1999). Riverine systems are connected in a dendritic pattern where distances between sites are defined by the length of the riverine path and not by simple linear distance, and this geometrical arrangement also determines species distribution (Chaput-Bardy, Fleurant, Lemaire, & Secondi, 2009). Thus, stream hierarchy and gradient, as well as riverine distances may influence gene flow among riverine populations (Perrier, Guyomard, Bagliniere, & Evanno, 2011).

Genetic diversity is related to the complexity of riverine systems (i.e. drastic topographic changes and hydroclimatic fluctuations). For example, topographic gradients (slope and waterfalls) can prevent dispersal (Narum, Zendt, Graves, & Sharp, 2008). Also, temperature fluctuations and flooding events can cause variation in population sizes (Hand et al., 2016; Wenger et al., 2011). Similarly, anthropogenic factors such as land-use changes (e.g. construction of dams or culverts) often increase aquatic habitat fragmentation, reducing functional connectivity among populations and increasing chances of local and system-wide extinction (Tortorotot, Perrier, Bergeron, & Bernatchez, 2014).

Additionally, the introduction of exotic species could have detrimental effects in native populations by interspecific competition, predation, diseases, parasites and the focus of our work here

genetic introgression (Penaluna et al., 2016). One of the most negative impacts of genetic introgression is the corruption of native genomes that have evolved over thousands of years resulting in loss of adaptive potential of local populations (Muhlfeld et al., 2009).

The endemic Mexican golden trout (*Oncorhynchus chrysogaster*) inhabits the Río Fuerte, Río Sinaloa and Río Culiacán basins in the Sierra Madre Occidental, representing one of the southernmost distributed salmonid species and the only one protected by Mexican regulations (Hendrickson et al., 2006). Its habitat is restricted to the highest parts of the Sierra Madre Occidental Mountains. Thus, these populations are endangered by anthropogenic and environmental factors, such as changing land use and climate, as well as the introduction of the exotic rainbow trout (*Oncorhynchus mykiss*) for aquaculture (Hendrickson et al., 2006). Recent studies found a cryptic genetic structure in *O. chrysogaster*: different genetic clusters, with some populations introgressed by cultured rainbow trout (Abadía-Cardoso, Garza, Mayden, & García-De León, 2015; Escalante et al., 2014). However, all previously published studies have used data from only a few locations and sample designs not adjusted for environmental gradients. Consequently, the knowledge of the real effect of landscape on *O. chrysogaster* population genetic structure and its relationship with introgressed genetics is still missing.

The genetic processes occurring in riverine landscapes (herein riverscapes) are highly influenced by the landscape through which they flow and their understanding requires a multi-scale approach due to the large amount of life-history traits occurring there (Fausch, Torgersen, Baxter, & Li, 2002). Combining species distribution models and landscape genetics simulations is one way to help improve our understanding of the complex effect of riverscapes on genetic processes (Scoble & Lowe, 2010; Scribner et al., 2016). Furthermore, simulation methods in landscape genetics have been used successfully to explain processes, such as migration, selection, mutation, recombination and drift in heterogeneous habitats (Andrello & Manel, 2015; Hoban, Gaggiotti, & Bertorelle, 2013; Landguth & Cushman, 2010). However, few simulation studies have been conducted in riverscape genetics (Table 1), even though such studies can help to disentangle process shaping patterns (Cushman, 2014). In this work, we aimed to investigate the effect of riverscape complexity and rainbow trout introgression on population structure of *O. chrysogaster*, addressing the following questions: (1) Do riverscapes drive *O. chrysogaster* genetic structure? and (2) Is genetic introgression influencing current *O. chrysogaster* genetic structure? To address these questions, we use an integrative approach combining species distribution modelling, empirically derived riverscape genetics simulations and riverscape genetics analyses. We discuss how data simulated under different demographic and genetic (herein defined as demogenetic; Landguth et al., 2014) scenarios fit with trout life-history traits in complex environments, and how this can help to interpret the process behind the observed genetic structure.

TABLE 1 Examples of studies applying simulations on riverscape genetics

Species	Location	Simulated processes	Main results	Reference
Westslope cutthroat trout <i>Oncorhynchus clarkii lewisi</i>	Akokala Creek in USA	Barriers driving demogenetic connectivity	Barriers placed at headwater streams caused loss of genetic diversity	Muhlfeld et al. (2012)
Electric fish <i>Steatogenys elegans</i>	Amazon and Negro Rivers in South America	Genetic structure and spatial selection	Difference in neutral and selection-driven genetic structure influenced by the spatial selection gradient	Cooke, Landguth, and Beheregaray (2014)
Bull trout <i>Salvelinus confluentus</i>	Upper Flathead River in Canada and USA	Stream resistance driving demogenetic connectivity	Riverscape resistance causing suitable habitat fragmentation and loss of genetic diversity	Landguth et al. (2014)
Westslope cutthroat trout <i>Oncorhynchus clarkii lewisi</i>	Washington, USA	Managed removal of non-native species and influence of barriers in genetic exchange	Inability of exotic populations to recolonize patches and lower genetic exchange when barriers are added	Landguth et al. (2016)
Freshwater mussel <i>Cumbrlandia monodonta</i>	Mississippi River in USA	Effects of climate change in demogenetic connectivity	Climate change will significantly reduce population connectivity and genetic diversity	Inoue and Berg (2017)

2 | MATERIALS AND METHODS

2.1 | Study area

This study focused on the *O. chrysogaster* distribution area described by Hendrickson et al. (2002), located at upstream waters in altitudes higher than 1,500 m at Río Fuerte, Río Sinaloa and Río Culiacán basins, Mexico. This zone has complex and abrupt topography with a river network mostly corresponding to 1, 2 and 3 stream order values, an average altitude of 2,107 m, an annual mean temperature of 14.4°C and an annual precipitation mean of 911 mm (Figure 1). The invasive rainbow trout was introduced to this area (with aquaculture purposes) in the 1860s (Contreras-Balderas & Escalante-Cavazos, 1984).

2.2 | Ecological modelling and empirical data

2.2.1 | Riverscape data

Initially, slope and aspect models were produced from Aster Digital Elevation Models (DEM) with 30 m resolution, downloaded from the Japanese Space Systems website (<http://gdem.ersdac.jpacesystems.or.jp/>). From this DEM, we defined the study area based on basin delineation. Then, a hydrologic model including the drainage network, stream orders, drainage density and stream length was created. Geological data (rock types) at 1: 250,000 scale were downloaded from the Mexican Institute of Statistics and Geography (<http://www.inegi.gob.mx>). We used 19 bioclimatic variables at 1 km resolution from the WorldClim website (<http://www.worldclim.org>). Lastly, we included 26 riverscape variables (see Appendix S1), which were geometrically corrected to 30 m to fit the DEM resolution. All

spatial analyses were conducted in ArcGIS 10.2 (ESRI, 2013; Table S1.1).

2.2.2 | Species distribution modelling

Oncorhynchus chrysogaster distribution area was modelled using MAXENT (MAXENT 3.33k software; Phillips, Anderson, & Schapire, 2006). We calculated a correlation matrix for each raster layer (Table S1.2) and riverscape variables were discarded based on a correlation cut-off of $r \geq 0.5$ to avoid collinearity. With the selected environmental information, MAXENT predicts the occurrence of the species using a maximum entropy algorithm that accounts for the properties of the reference locations (occurrence records). For this study, 45 occurrences from 1952 to 2014 (all the historical records for *O. chrysogaster* till 2014) were employed as presence data, and the 26 aforementioned riverscape variables were used to run an initial MAXENT model. Variables with a weak influence (AUC with jackknife test <0.7) on *O. chrysogaster* distribution were then discarded. The area under curve (AUC) score was used to assess model accuracy (Guisan & Zimmermann, 2000; Manel et al., 1999). The resulting reduced model allowed the initialization of starting locations for the simulation program, as well as helping to inform riverscape resistance surface creation.

2.2.3 | Description of empirical genetic data

The empirical genetic data consisted of 189 wild *O. chrysogaster* individuals from seven sampling sites and 76 *O. mykiss* individuals from four aquaculture farms genotyped using 11 microsatellites. Native trout showed unbiased expected and observed heterozygosity means of 0.466 and 0.566, respectively. Bayesian clustering analyses

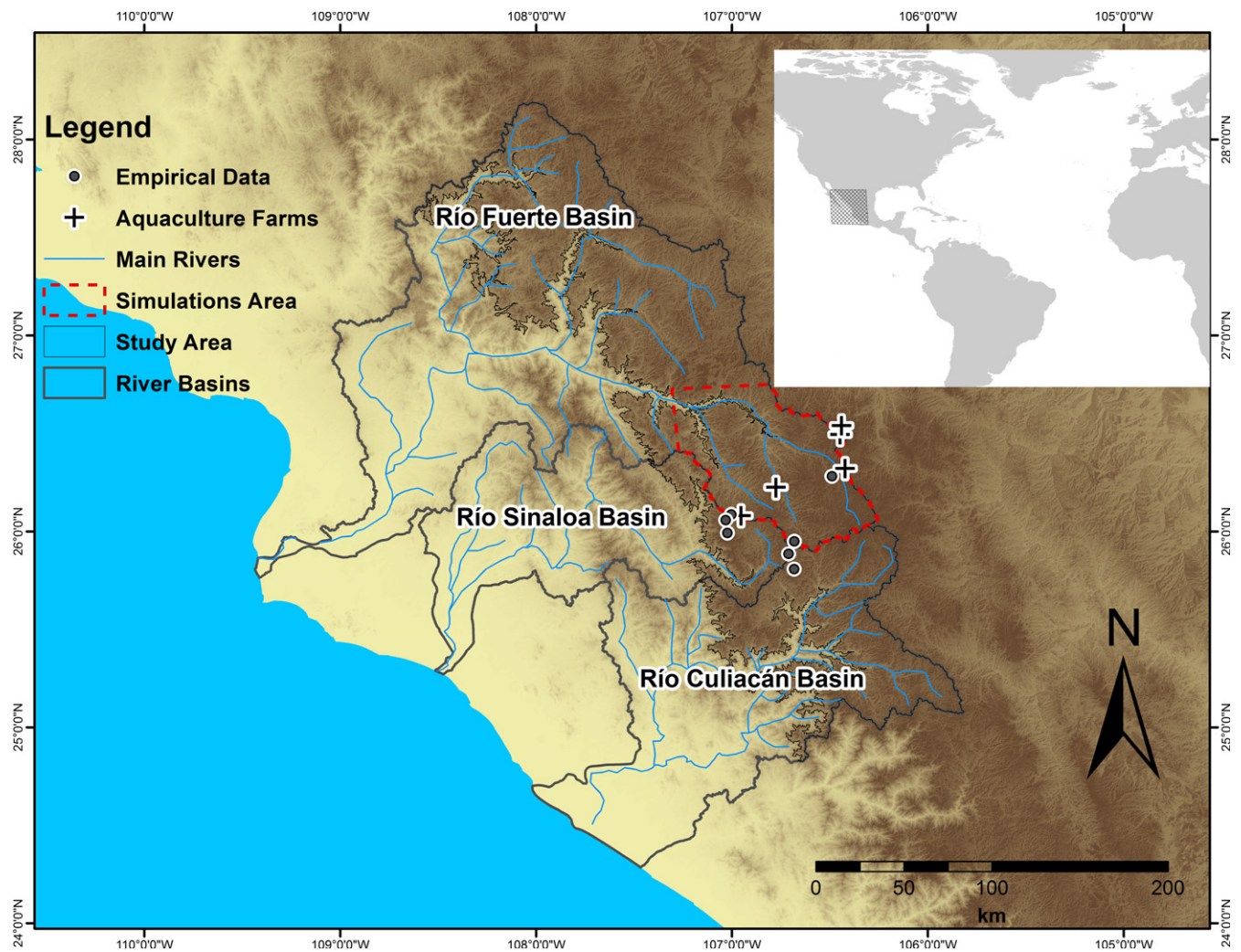


FIGURE 1 Study Area: Río Fuerte, Río Culiacán and Río Sinaloa basins over 1,500 m above the sea level in northwest Mexico. Dots indicate the location of empirical data sampling sites; crosses indicate the location of aquaculture farms, and the simulated study system is surrounded by red lines

defined three *O. chrysogaster* genetic clusters and one aquaculture cluster for that dataset. Native genetic clusters presented low degrees of introgression from the aquaculture cluster and slight levels of admixture with other native clusters (see Escalante et al., 2016 for further information). The allele frequencies from individuals of the three native clusters and the aquaculture cluster were used to initialize the simulations.

2.3 | Riverscape genetics simulations

2.3.1 | Riverscape demogenetic program

Riverscape influences on genetic exchanges and introgression processes in *O. chrysogaster* populations were simulated using the newly developed riverscape genetics program, CDMetaPOP (Landguth, Bearlin, Day, & Dunham, 2016). This approach models demogenetic processes (e.g. growth, movement, mating, genetic connectivity) for individuals located at different geographical locations (patches). In

order to distinguish between our two species, we initialized patches with individuals that had allele frequencies matching each species. Inside patches, individuals undergo migration, reproduction and mortality, and the subsequent genetic processes over a user-specified given time step at the individual level. Further explanation about the theoretical life cycle is found in the software manual (<http://www.github.com/computationalecologylab/CDMetaPOP/>). Details on the major parameters and processes as it relates to our research questions are described below and more information is available in Appendix S2.

2.3.2 | Patches and carrying capacity

Due to the size of the study area, we focused our simulations only on the highest parts of the south of Río Fuerte Basin (Figure 1). This area also has rainbow trout aquaculture farm locations. Figure 2 shows the patch locations delineated as natal and migratory (i.e. overwintering/foraging). We applied a *O. chrysogaster*'s species distribution model probability of occurrence cut-off of 0.8 to initialize



249 patches considered natal spawning ground locations, and 1,692 sites were considered migratory patches. We defined the starting source locations for rainbow trout from the geographic location of five aquaculture farms obtained in field surveys and through government and academic agencies.

For *O. chrysogaster* patches (both natal and migratory), carrying capacities ranged from 69 to 255 individuals and initial number of individuals ranged from 31 to 137. These values were derived using data on native coastal Nelson trout (*Oncorhynchus mykiss nelsoni*; Ruiz-Campos & Pister, 1995) and field observations. For rainbow trout source locations, we fixed a carrying capacity of 10,000 with 9,200 starting individuals. See Appendix S2 for more details on patch initiation.

2.3.3 | Genetic data

The genotype for each individual was initialized considering for native and exotic trout and we did not consider allele frequencies of admixed individuals either with *O. chrysogaster* or rainbow trout in order to test our hypothesis with non-introgressed genetic clusters. We discarded individuals based on an admixture coefficient cut-off ≥ 0.001 . Thus, *O. chrysogaster* allele frequencies ($n = 57$) from the three empirical genetic clusters were used to simulate native trout at three different subregions, depending on their geographical location and their proximity to the predefined genetic clusters. The specific empirical allele frequencies for the aquaculture cluster ($n = 26$) were used to mimic rainbow trout at aquaculture farms (Appendix S2). The simulation program then simulated genetic exchange following Mendelian inheritance and specified species-specific movement and life-history strategies described below.

2.3.4 | Age classes, migration, maturation and fecundity

As we lacked the necessary empirical data to inform all model variables, we defined them using a similar system, westslope cutthroat trout (i.e. WCTM model; Landguth et al., 2016; Appendix S2). The model was initialized at time = 0 with eight age classes. Migration rates started from 0 at age 0 with an annual increment of 0.1 until age 4, and a straying probability of .05 was assigned to all age classes. Maturation rates and the fecundity model were taken from the WCTM with the exception of rainbow trout individuals, where maturation rates were slightly modified based on data published for *O. mykiss* farming (Dirección General de Acuicultura, 1982) to create earlier maturing individuals (Appendix S2).

2.3.5 | Dispersal

Resistance surfaces quantified the influence of landscape features on functional population connectivity (Spear, Balkenhol, Fortin, McRae, & Scribner, 2010). Thus, a riverscape resistance surface was created where each pixel represented the cost of crossing each location using four environmental variables: altitude, slope, mean temperature of the warmest quarter and stream order. The four environmental

variables were scaled and processed in ArcGIS (Appendix S3). These variables were chosen based on field surveys (M.A.E. personal observations) and also the response curves at the MAXENT model (Figure 4). In other studies, temperature increase, abrupt slopes, altitudinal gradient and increase of stream order were suggested as inducing riverscape resistance for trout movement (Hendrickson et al., 2006; Meeuwig, Guy, Kalinowski, & Fredenberg, 2010).

To quantify landscape fragmentation and connectivity between populations, the least accumulative cost distance for each patch to the nearest source (least cost distance) was calculated accounting for riverine distance and riverscape resistance surfaces using the 'gdistance' package (van Etten, 2012) in R 3.2.5 (R Development Core Team, 2016) to generate matrices of connectivity values among all patches. This package simulates the potential movement of species in a spatially structured landscape, linking different dispersal functions and connectivity thresholds based on Dijkstra's shortest path algorithm (Dijkstra, 1959). We generated four matrices under different movement hypotheses: (Matrix I) hypothesizing movement between native patches with isolation by riverine distance, using only the length of the river network; (Matrix II) hypothesizing movement between native patches by riverscape resistance, using the resistance surface to subsequently analyse with 33% of maximum resistance threshold (preventing dispersal after accumulating the 33% of the total resistance); (Matrix III) hypothesizing movement between native and aquaculture patches with isolation by riverine distance, using only the length of the river network; and (Matrix IV) hypothesizing movement between native and aquaculture patches by riverscape resistance, using the resistance surface to subsequently analyse with 33% of maximum resistance threshold (preventing dispersal after accumulating the 33% of the total resistance). For further information, see Appendices S2 and S3.

2.3.6 | Simulation scenarios

Given the approximate date of introduction of rainbow trout into the area, we performed simulations for 150 years. Four different demogenetic scenarios covering a wide spectrum of factors that could modify the spatial genetic structure were considered: (I) Isolation by riverine distance without exotic introgression; (II) Isolation by riverscape resistance without exotic introgression; (III) Isolation by riverine distance with exotic introgression; and (IV) Isolation by riverscape resistance with exotic introgression (Table 2 and Appendix S2). Additional sensitivity analyses were conducted, based on Scenario II parameters (Appendix S4), to test how genetic diversity and structure change under different resistance surfaces. For all simulations, we ran three replicates to optimize computational time.

2.4 | Analyses of simulation outputs

2.4.1 | Genetic diversity

Observed heterozygosity (H_o), unbiased expected heterozygosity (H_e) and deviations from Hardy–Weinberg equilibrium were estimated

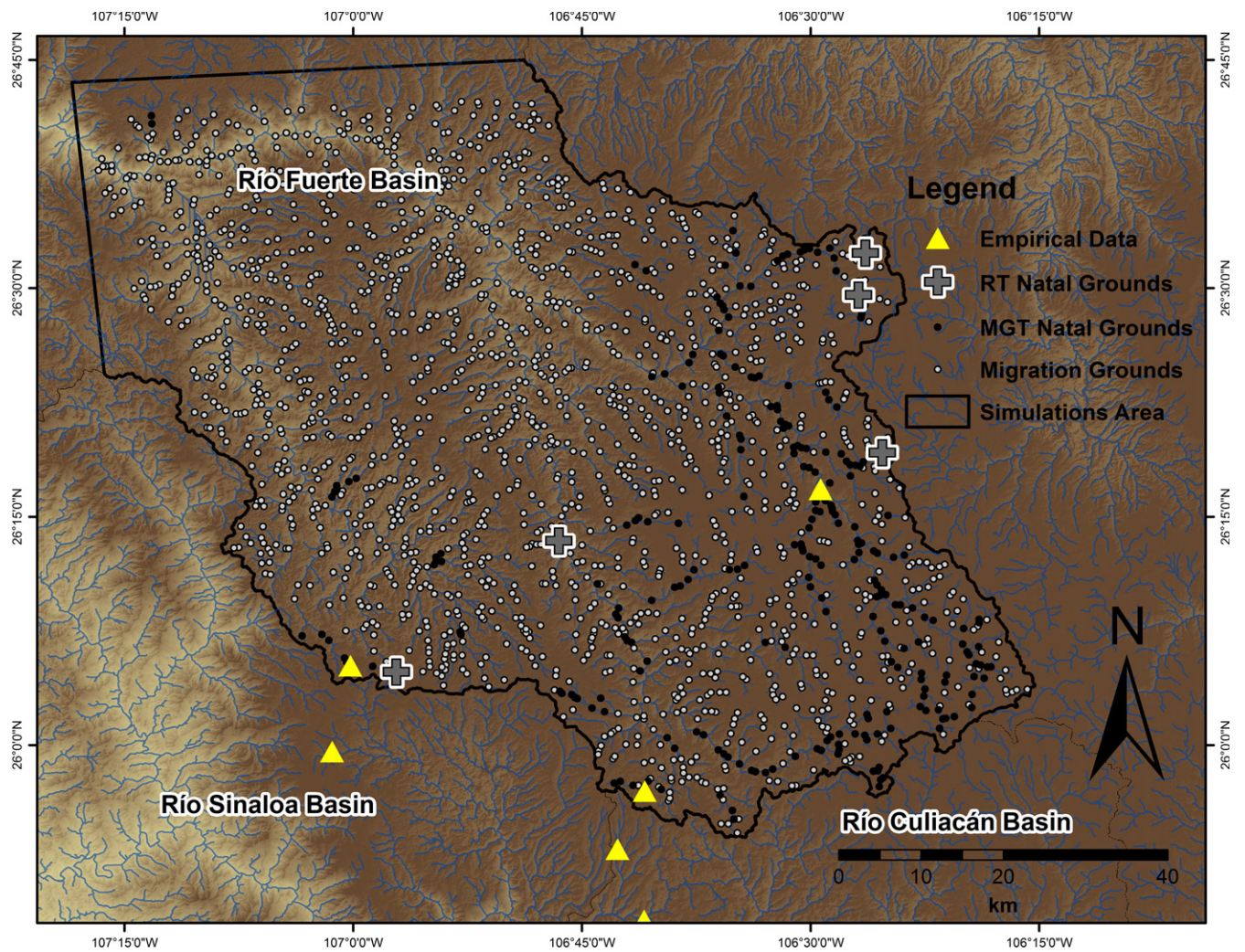


FIGURE 2 Initial simulated patches at the south of Río Fuerte Basin: yellow triangles indicate the location of empirical data sampling sites, black dots indicate Mexican golden trout (MGT) natal grounds, crosses indicate cultured rainbow trout (RT) natal grounds, grey dots indicate migration ground and the simulated study system is surrounded by the black polygon

using GENALEX 6.5 (Peakall & Smouse, 2006). To validate significant differences between scenarios allowing comparisons with empirical data, pairwise Student's *t* test (Student, 1908) was performed among scenarios for the H_o and H_e values of year 150 (last replicate) using the Excel Macro XLSTAT 2017.1 (Adinsoft, 2017). Additional estimates of allelic richness (AR) accounting for rarefaction were conducted in HP-Rare (Kalinowski, 2005; only last replicate in year 150) considering a minimum of 6 (Scenario I), 14 (Scenario II and III) and 18 (Scenario IV) individuals.

2.4.2 | Population structure

Genetic distances were first estimated among all patches using pairwise F_{ST} coefficients in GENALEX. The spatial genetic structure for year 150 (in the last replicate) of all scenarios was assessed using spatial principal component analysis (sPCA) available in the R package 'ade genet' (Jombart, 2008; Narum et al., 2008). This method combines variability and spatial autocorrelation of allele frequencies

to generate synthetic principal components describing spatial genetic structure. To analyse the effect of exotic introgression on population genetic structure, principal component analysis (PCA) included in ADEGENET was used to compare genetic variability on allele frequencies for the last replicate at the first year (before migration started; generation = 1) and year 150 across the riverscape resistance simulation scenarios (II and IV).

2.4.3 | Riverscape genetics

To assess the spatial influence of the two movement surfaces (riverine distance versus riverscape resistance), we used Mantel tests (Mantel, 1967) implemented using XLSTAT on each riverine/ riverscape surface scenario and its corresponding genetic distance matrix (F_{ST} calculated in GENALEX). Those analyses were applied between F_{ST} coefficients for the year 150 (last replicate) of all scenarios. All tests were performed under 10,000 permutations assuming no correlation.



TABLE 2 Main model parameters for each *Oncorhynchus chrysogaster* simulated scenario (I–IV) at the south of Río Fuerte Basin. West slope cutthroat model (WCTM)

Scenario	Scenario's name	Age classes, migration, maturation and fecundity	Distance/resistance matrix	Initial number of individuals per patch	Carrying capacity per patch	Dispersal movement thresholds of riverscape distance/resistance
I	Isolation by riverine distance without exotic introgression	WCTM (Landguth et al., 2016)	Matrix I	From 31 to 137	From 69 to 255	—
II	Isolation by riverscape resistance without exotic introgression	WCTM (Landguth et al., 2016)	Matrix II	From 31 to 137	From 69 to 255	—
III	Isolation by riverine distance with exotic introgression	WCTM (Landguth et al., 2016)/ Dirección General de Acuacultura (1982)	Matrix III	From 31 to 137 for native patches 9,200 for exotic patches	From 69 to 255 for native patches 10,000 for exotic patches	33%
IV	Isolation by riverscape resistance with exotic introgression	WCTM (Landguth et al., 2016)/ Dirección General de Acuacultura (1982)	Matrix IV	From 31 to 137 for native patches 9,200 for exotic patches	From 69 to 255 for native patches 10,000 for exotic patches	33%

3 | RESULTS

3.1 | Species distribution modelling

The MAXENT-based species distribution model detected three weakly correlated variables with robust influence on *O. chrysogaster* presence (Table S1.1): altitude, precipitation of the driest month and mean temperature of the warmest quarter. The highest probabilities of *O. chrysogaster* occurrence were located at the east and south of Río Fuerte Basin, east of Río Culiacán Basin and the limits between Río Fuerte and Río Sinaloa basins and Río Culiacán and Río Sinaloa basins (Figure 3). The AUC score was high (0.87) and the model successfully predicted 95.3% of the species records. Thus, we estimated the best conditions for the species' occurrence at places with a precipitation of the driest month close to 22 mm, altitudes around 2,600 m and a mean temperature of the warmest quarter lower than 15°C (Figure 4). The three subregions for the simulations were separated by ruptures in the MAXENT model represented by low species occurrence probabilities, having similar environmental conditions but isolated by the heterogeneous riverscape.

3.2 | Analyses of simulation outputs

3.2.1 | Genetic diversity

The highest heterozygosity values were reached by Scenario III (Isolation by riverine distance with exotic introgression) with means of $H_o = 0.788$ and $H_e = 0.746$ in year 150 of the last replicate. The lowest values that were most similar to the empirical data were found at Scenario II (Isolation by riverscape resistance without exotic

introgression) with means of $H_o = 0.634$ and $H_e = 0.605$ for the last replicate of Year 150 (Table 3; see Appendix S5 for values plotted through the time for all replicates), those values also matched the best sensitivity analyses (Appendix S4). Mean AR ranged from 2.72 (Scenario I) to 5.60 (Scenario IV). The proportion of loci under Hardy–Weinberg disequilibrium ($p < .05$) ranged in year 150 of the last replicate from 61.7% (Scenario II) to 80.9% (Scenario III; Table 3). Furthermore, the t test revealed significant differences ($p < .0001$) for heterozygosity values among all scenarios. Scenarios III and IV including exotic introgression exhibited the higher genetic diversity values. However, the one including only riverine distance showed higher scores, indicating a stronger introgression pattern when riverscape resistance was excluded.

3.2.2 | Population structure

Scenarios simulated under riverscape resistance showed higher degrees of genetic differentiation. The highest global F_{ST} values between populations at year 150 observed in Scenario II (e.g. 0.190 in last replicate) were the most similar to the empirical data (0.20), while Scenario III obtained the lowest values (e.g. 0.060 in last replicate). Table 3 shows only the values for the last replicate at year 150, and Appendix S5 shows all replicate values plotted through time.

The sPCA performed for scenarios I and III did not detect genetic clusters (results not shown). However, for scenarios under riverscape resistance (II and IV) and sensitivity analyses (Appendix S4), sPCAs detected three different clusters separated into three subregions corresponding to the main river branches (west, centre and east) and separated with high riverscape resistance agreeing with empirical

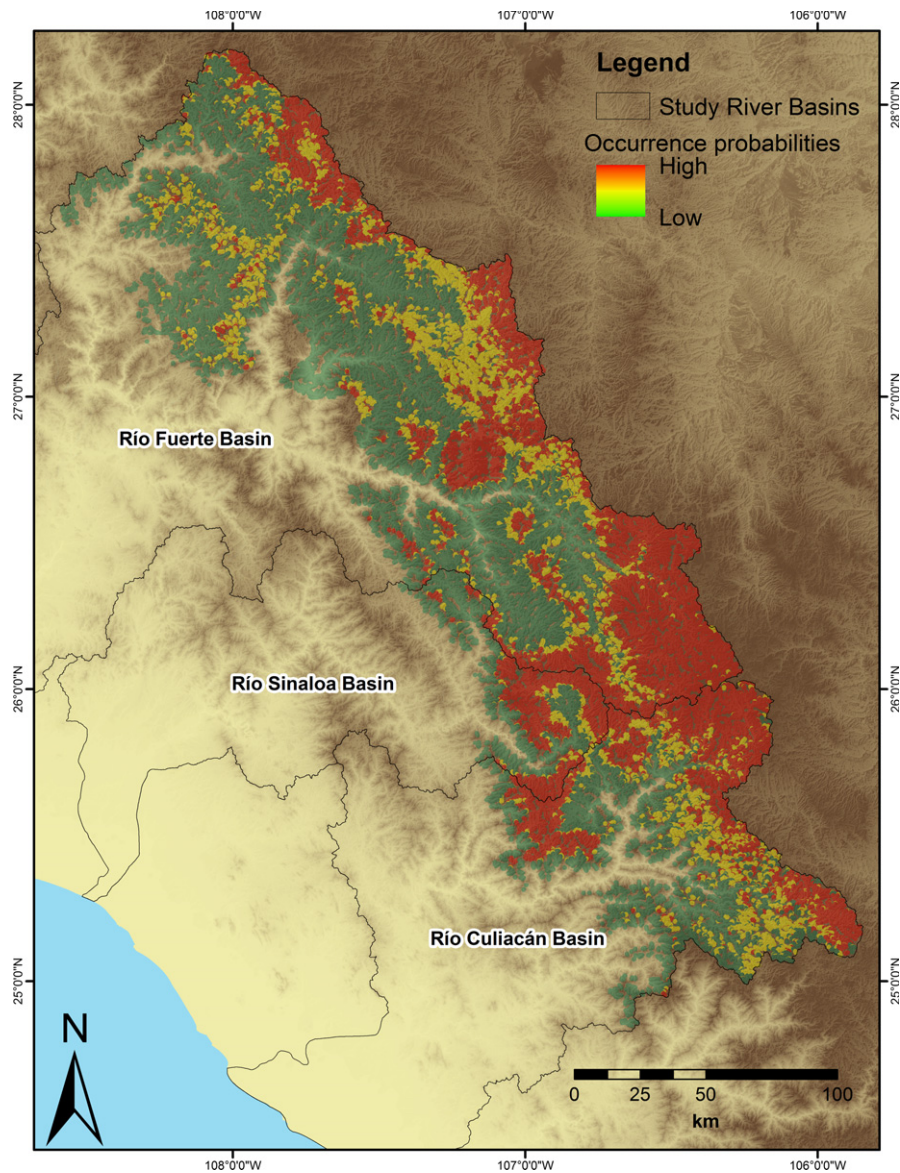


FIGURE 3 *Oncorhynchus chrysogaster* occurrence probabilities derived from the species distribution modelling in MAXENT at Río Fuerte, Río Sinaloa and Río Culiacán. Zones with high occurrence probabilities are shown in red, zones with medium probabilities in yellow and zones with low probabilities in green

data. In those analyses, the first two eigenvalues of the sPCAs explained almost 100% of the variance in the data and were retained (Figure 5).

PCA results from examining introgression for year 1 of Scenario II detected three genetic clusters of *O. chrysogaster* (west, centre and east) agreeing with the sPCAs and empirical data (Figure 6a). For the first generation of Scenario IV, we identified the same three genetic clusters of *O. chrysogaster* clearly separated from rainbow trout (Figure 6b). For year 150 of Scenario II, the three *O. chrysogaster* clusters were still present, with the west cluster clearly separated from the rest and east and centre clusters sharing some admixed individuals (Figure 6c). Furthermore, for year 150 of Scenario IV, individuals from aquaculture origins were dispersed in the three *O. chrysogaster* clusters. Additionally, *O. chrysogaster* clusters

were less separated in comparison with analyses without introgression; this proximity could be an effect of genetic admixture with rainbow trout located at the three subregions (Figure 6d). The later pattern is less consistent with the empirical data in relation to the other outputs.

3.2.3 | Riverscape genetics

All Mantel tests were highly statistically significant ($p < .0001$). R -values were substantial only for Scenarios II (0.906) and IV (0.675), simulated under riverscape resistance, while scenarios considering just riverine distance (I and III) obtained values near zero. Those results are consistent with the sPCAs, corroborating the strong influence of the riverscape structure characterized by long riverine

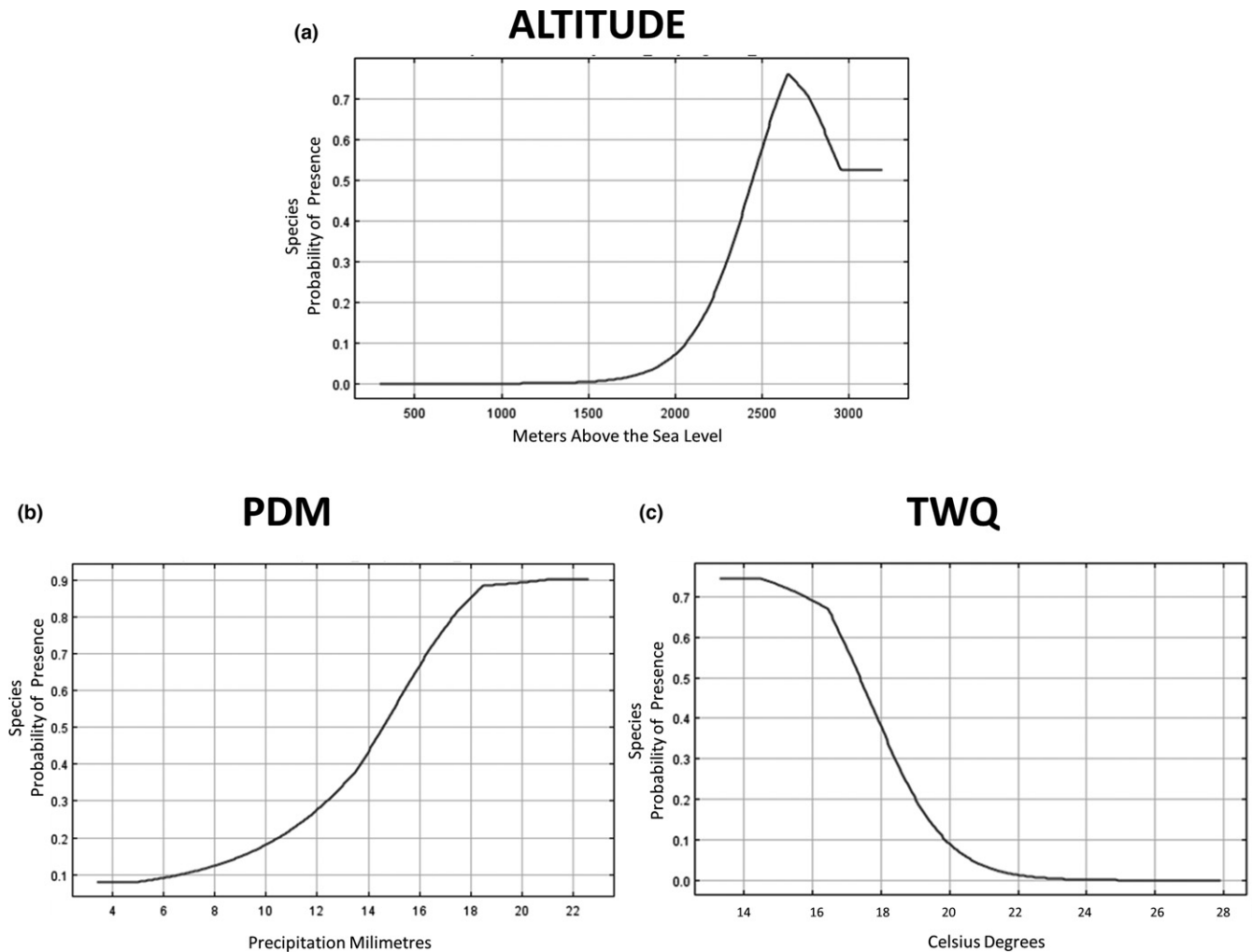


FIGURE 4 Environmental functions of the species distribution modelling in MAXENT for *Oncorhynchus chrysogaster*. The curves predict the changes of species distribution as a function of each environmental variable: altitude (a), precipitation of the driest month (PDM; b) and mean temperature of the warmest quarter (TWQ; c)

distances, abrupt slopes, stream order changes as well as elevation and temperature gradients on the genetic differentiation of *O. chrysogaster*.

4 | DISCUSSION

Our main findings from this comparative study showed that the simulated genetic structure in the scenario with riverscape resistance without exotic introgression strongly matched the observed genetic structure. The comparison of model scenarios suggests a strong effect of riverscape resistance on the species genetic structure. Moreover, we found evidence that riverscape resistance is acting as a boundary against exotic introgression.

Our most plausible scenario (Scenario II) based on mountaintop trout life-history knowledge, highlighted a cryptic genetic structure reaching the highest global F_{ST} values among scenarios. Such patterns of riverscape influence on spatial genetic structure are

expected in salmonids inhabiting heterogeneous habitats (Timm, Hallerman, Andrew Dolloff, Hudy, & Kolka, 2016), as is the case in our study system. The three well-defined genetic clusters separated by strong riverscape resistance corresponded to the main river branches at the south of Río Fuerte Basin. The scenario simulated with exotic introgression and riverscape resistance (IV) was the second most accurate presenting a similar cryptic spatial genetic structure, confirming the importance of riverscape to explain this genetic variation. In this scenario, genetic introgression was considerably less strong, indicating that a complex riverscape could act as a boundary against introgression. This suggests that riverscape is the main driver on the species genetic structure, most likely due to the influence of riverscape features on dispersal movements. Additionally, Mantel tests detected a strong significant correlation among riverscape resistance and genetic distances. Moreover, the genetic discontinuities matched riverscapes with higher resistance represented by increased temperature, decreased altitude, steeper slopes and higher stream order. Thus, we suggest that the genetic structure of *O.*

TABLE 3 Genetic diversity values and F_{ST} coefficients for the last replicate of the 150th year simulated under the scenarios considering *Oncorhynchus chrysogaster* (I and II) and both *O. chrysogaster* and cultured rainbow trout (III and IV) at the south of Río Fuerte Basin, and native *O. chrysogaster* empirical data (NEP): Sample size (n), allelic richness (AR), observed heterozygosity (H_o), unbiased expected heterozygosity (H_e), global F_{ST} between all populations (F_{ST}) and proportion of loci under Hardy–Weinberg disequilibrium (%HW). The mean (M) and standard error (SE) are shown for H_o , H_e and F_{ST}

Scenario	n	AR	H_o	H_e	F_{ST}	%HW
I	39,490	M 3.24	M 0.726 SE 0.004	M 0.682 SE 0.003	M 0.068 SE 0.001	75.10
II	31,350	M 3.98	M 0.634 SE 0.005	M 0.605 SE 0.004	M 0.190 SE 0.029	61.70
III	61,426	M 5.39	M 0.788 SE 0.003	M 0.746 SE 0.003	M 0.060 SE 0.001	80.90
IV	39,927	M 5.60	M 0.773 SE 0.003	M 0.731 SE 0.003	M 0.088 SE 0.004	77.50
NEP	179	M 2.72	M 0.466 SE 0.046	M 0.566 SE 0.045	M 0.200 SE 0.050	66.00

chrysogaster is most likely shaped by the complex riverscape structure rather than just riverine distance. These findings illustrate the need to consider landscape characterization as an important explanatory variable for spatial genetic structure in riverscape systems (Hand et al., 2016; Meeuwig et al., 2010).

Riverscape variables used in the MAXENT model were also included in the resistance surfaces, with two additional variables related to dispersal (slope and stream order). Thus, we found that similar riverscape conditions drive species occurrence, as well as demographic and genetic processes, which is not always the case (Dellicour et al., 2016; Scoble & Lowe, 2010). The strong influence of temperature of the warmest quarter, precipitation of the driest month and altitude on *O. chrysogaster* occurrence was also found in similar studies of other trout species from North America (Hendrickson et al., 2006; Wenger et al., 2011). These studies showed a positive effect of temperature decrease, as well as altitude and precipitation increase on trout occurrence, while the simulations confirmed that those variables are also shaping the genetic structure of *O. chrysogaster*.

Additionally, the comparison of genetic variation among scenarios permits an evaluation of the potential effects of introgression and riverscape on native populations. The highest genetic diversity values among scenarios and lowest F_{ST} values were found for simulations of exotic introgression without riverscape resistance. Those values can be explained by genetic admixture between exotic and native individuals, as it is expected in native populations with frequent introgressions (Berrebi, Povz, Jesensek, Cattaneo-Berrebi, & Crivelli, 2000). This phenomenon has been observed previously for Atlantic salmon, where populations subjected to high genetic introgression obtained lower F_{ST} values compared with slightly or non-introgressed populations (Perrier et al., 2011). Similarly, a simulation study on *Salmo salar* found less genetic introgression from aquaculture stocks when populations were exposed to lower dispersal levels (Perrier, Baglinière, & Evanno, 2013). Additional observations including *O. chrysogaster* and undescribed trout from Sierra Madre

Occidental (Abadía-Cardoso et al., 2015; Escalante et al., 2014) detected the highest genetic diversity values for introgressed populations located near aquaculture farms.

Furthermore, among both scenarios with exotic introgression, the one with riverscape resistance (Scenario IV) showed lower genetic diversity values and higher F_{ST} values. Those results support the hypothesis about the influence of riverscape structure on trout dispersal. Taking into account that *O. chrysogaster* habitat is characterized by abrupt hydroclimatic and topographic gradients, we believe that riverscape heterogeneity is mitigating introgression processes as consequence of reduced functional connectivity between rainbow trout sources and some native populations.

4.1 | Comparisons among simulations and empirical data

The most plausible scenario (Scenario II) was also the most similar in relation to empirical data, exhibiting a similar global F_{ST} and heterozygosity. sPCAs defined the three genetic clusters observed in the empirical data for both scenarios simulated under riverscape resistance (II and IV). However, PCAs highlighted a more realistic pattern for Scenario II with the three clusters clearly defined and lower genetic admixture among them. Previous empirical analyses have suggested low introgression from exotic trout for *O. chrysogaster* for the study area (Escalante et al., 2014, 2016), which is consistent with the outcomes of Scenario II.

4.2 | Limitations and uncertainties

This work has presented a novel integrative approach through the combination of species distribution and demogenetic, spatially explicit simulation models, detecting hydroclimatic variables with high influence on the distribution of an endemic salmonid and the effect of riverscape resistance on spatial genetic structure and exotic introgression. However, although Scenario II was the most realistic, we

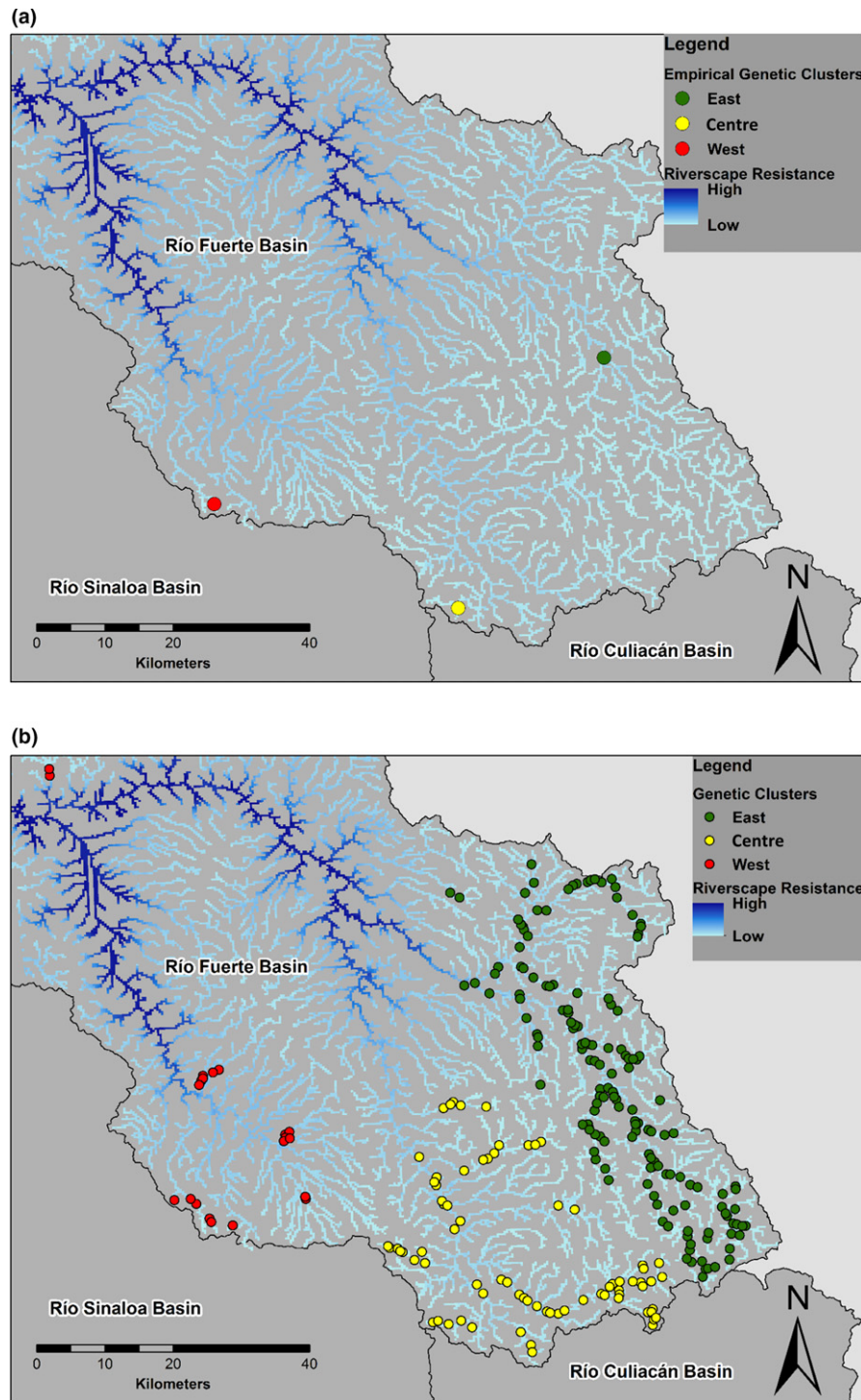


FIGURE 5 Geographic distribution of the empirical *Oncorhynchus chrysogaster* genetic clusters (a) and spatial principal component analysis (sPCA) outputted genetic clusters for scenarios II and IV (b): east in green circles, centre in yellow circles and west in red circles

need to be cautious with some assumptions. Other studies, based on distinct molecular markers and sampling locations for *O. chrysogaster* at the south of Río Fuerte Basin, showed a different number of genetic clusters ranging from one to three (Abadía-Cardoso et al., 2015; Camarena-Rosales et al., 2008; Escalante et al., 2016). Thus,

the real population genetic structure of the species is not completely understood yet. Moreover, the lack of information about the biology of this species and the strength of aquaculture activities and escape rates from farms make simulating introgression conditions more challenging.

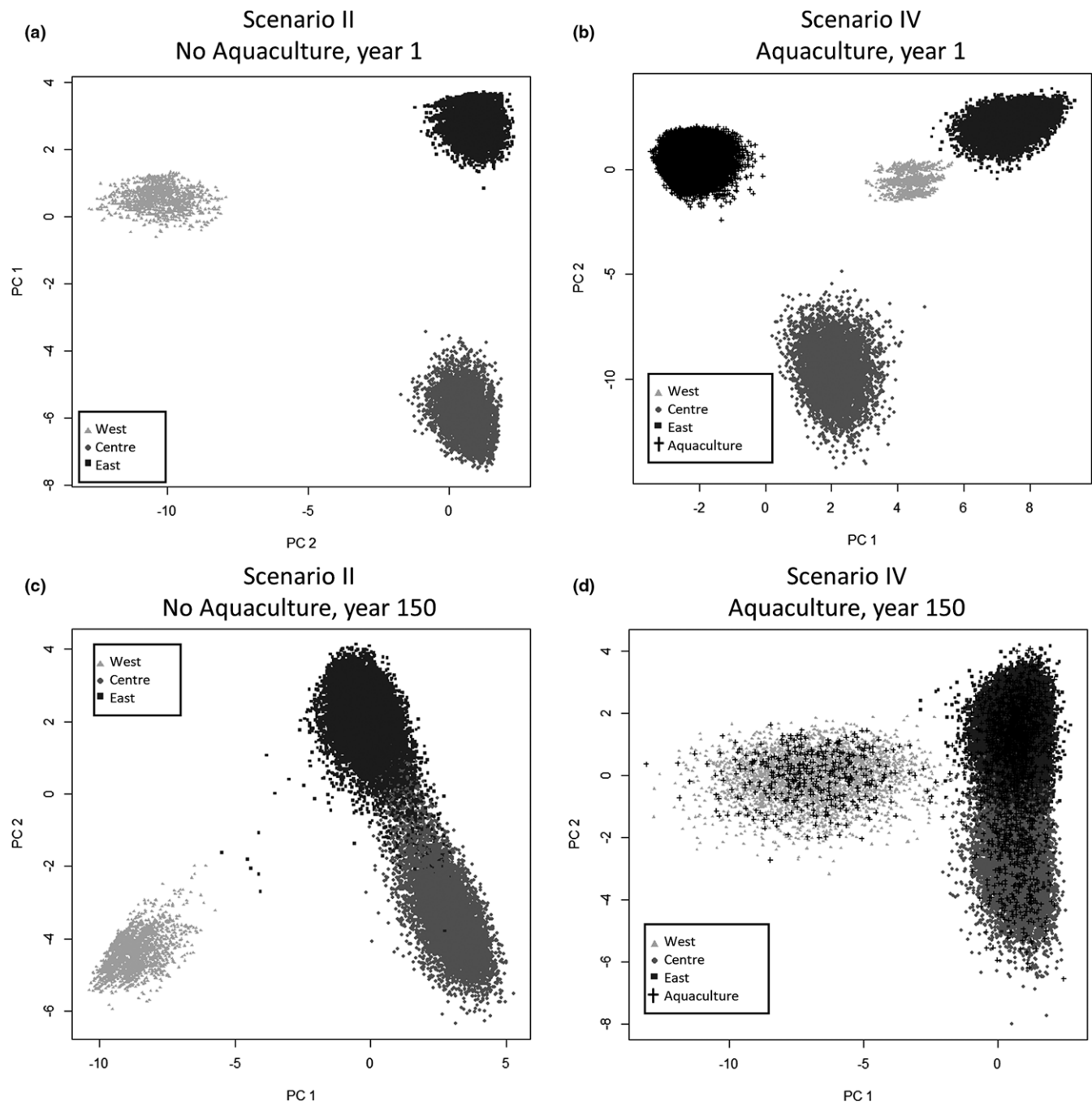


FIGURE 6 *Oncorhynchus chrysogaster* and cultured rainbow trout genetic clusters obtained from principal component analysis (PCA) on simulated data for Scenario II at generation 1 (a), Scenario IV at year 1 (b), Scenario II at year 150 (c) and Scenario IV at year 150 (d). The origin of each individual is represented by different figures: east in light grey triangles, centre in grey circles, west in dark grey squares and aquaculture in black crosses

4.3 | Implications for conservation

Although this study suggested low introgression from aquaculture trout in *O. chrysogaster*, native populations at aquaculture proximities are still at risk, as high introgression has been observed in non-described Mexican trout at the south of Sierra Madre Occidental (Escalante et al., 2014). It has been reported that invasions of salmonids from aquaculture origins could have harmful effects in native

populations putting them in risk of extinction (Penaluna et al., 2016). This issue highlights the need to gather more data to study this species that samples a wider geographic distribution, supported by a higher number of polymorphic molecular markers. Next generation sequencing opens new perspectives to generate large datasets of high resolution molecular markers (e.g. Elshire et al., 2011) and their use could aid understanding of the respective influence of landscape and exotic trout on the genetic structure of this salmonid, assisting

conservation strategies that take into account both evolutionary and ecological approaches.

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BIOSKETCH

Marco A. Escalante is broadly interested in landscape ecology, landscape genetics, G.I.S. and conservation of endangered species. He conducted this study as part of his PhD project at CEFE CNRS and CIBNOR.

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SUPPORTING INFORMATION

Additional Supporting Information may be found online in the supporting information tab for this article.

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